# Research Mapping of IIIT Allahabad, ABV-IIITM Gwalior and IIIT Bangalore, 2016–2026

DPCN 2026 Course Project · **Group 10** · Team Titan Mandal · Hrishikesh, Sarvesh, Shourya · Instructor: Chittaranjan Hens

| Resource | Link |
|---|---|
| Interactive dashboard (all panels, networks, findings) | dashboard.html / site/index.html, built by part4_dashboard.py (public URL in the README) |
| Code and documentation (README.md, DECISIONS.md) | https://github.com/sarvesh2005takbhate/dpcn_A2 |
| Streamlit dashboard (filters by institute / year / type) | streamlit run app.py (in the repository) |
| Submission files (to the instructor; not in the public repository) | group10_data.csv · citation_network.csv · authorship_network.csv · indian_connections.csv · group10_faculty.csv |

## Research question and key findings

**How do the research networks of the three IIITs compare, and how — and through whom — are the institutes connected?** Co-authorship edges = two authors on one paper (undirected, weighted by joint papers); citation edges = paper A lists paper B as a reference (directed).

- **Three institutional islands inside one connected component.** Only 138 of 8,667 co-author links (1.59%) join two different IIITs, although all three sit in one giant component of 3,265 authors.
- **The IIITA–ABV link is carried by people who moved, not by joint projects.** IIITA–ABV accounts for 116 of the 138 inter-institute links and 29 co-published papers, against 8 (IIITA–IIITB) and 1 (ABV–IIITB).
- **IIIT Bangalore hangs on eight people.** Every path from IIITB to IIITA or ABV passes through the same 8 authors (minimum node cut); IIITA–ABV needs 28.
- **Cross-institute reachability collapses when brokers are removed.** 64% of cross-institute author pairs are joined by some path. Removing the 1% of authors with the highest betweenness cuts this to 17%, and 2% to 0.7%.
- **The IIITs are connected indirectly, through the same IITs.** 69 Indian institutions co-author with all three IIITs; the strongest common partners are Indian Institute of Technology Kanpur and Indian Institute of Technology Roorkee.
- **Author disambiguation changes the inter-institution picture.** Merging authors only because their names match (the earlier rule) creates false bridges between institutes; the conservative rule removes them.
- **Output more than doubled, at the same pace everywhere.** Papers grew 11.9% / 11.1% / 11.7% a year (IIITA / ABV / IIITB, log-linear fit 2016–2025, all $p < 0.001$): IIITA 174→374, ABV 126→332, IIITB 77→184 papers.
- **Quality rose with quantity.** The Q1 share of journal papers rose from 30.2% to 44.2% at IIITA and from 26.2% to 40.8% at ABV between 2016–19 and 2022–25 ($\chi^2$, $p < 0.001$). IIITB already started high (42.5% → 54.6%, $p = 0.076$), so growth did not come at the cost of venue quality.
- **Raw citations exaggerate the gap between institutes.** IIITB's median paper has 2 citations vs 4 at IIITA, but 70% of its output is conference papers. Normalised for year, field and type, mean impact is 1.048 (ABV), 1.013 (IIITA) and 0.901 (IIITB, 95% CI 0.828–0.98): a modest, not a large, difference.
- **The 'international premium' is really a venue and team-size effect.** Internationally co-authored papers score 1.242 vs 0.941 for single-institute papers (Mann–Whitney $p < 0.001$). Once venue, type, team size, year and field are held fixed (OLS, $n = 5{,}087$), the international effect is +2% (n.s.), while a Q1 venue adds +137%, a conference venue -51%, and doubling the team +27%.
- **Citations follow collaboration, not topic overlap.** Institutes that co-author more with us also cite us more (Spearman $\rho = 0.669$ / 0.691 / 0.579, all $p < 0.001$). Co-author partners are about a third of the Indian institutes that cite us but supply 57.0–72.5% of those citations. Topic similarity does not predict who we collaborate with ($\rho = 0.102$, $p = 0.59$).

## 1. Data and institution overview

| Metric | IIIT Allahabad | ABV-IIITM Gwalior | IIIT Bangalore | All (unique) |
|---|---|---|---|---|
| Papers (journal / conference) | 2,946 (1,716 / 1,230) | 2,198 (1,373 / 825) | 1,437 (430 / 1,007) | 6,543 (3,501 / 3,042) |
| Q1 share of journal papers | 40.1% | 38.7% | 50% | 40.6% |
| CORE A*/A conference papers | 10 | 5 | 21 | 36 |
| Citations (median per paper) | 35,527 (4) | 31,586 (3) | 8,577 (2) | 75,408 (3) |
| Institutional h-index (2016–2026 papers) | 73 | 66 | 38 | 93 |
| Core faculty / affiliated authors | 75 / 1,831 | 46 / 1,182 | 40 / 1,064 | 160 / 4,041 |
| Median core-faculty h-index | 9 | 11 | 5 | 8 |
| Papers with another Indian institute / international | 41.3% / 14.8% | 52% / 13.6% | 34% / 17.9% | 42.9% / 15% |
| Cite an Indian institute / cited by one | 73.8% / 59% | 75.9% / 59.4% | 58.6% / 37.4% | 71.2% / 54.4% |

**Data.** OpenAlex works affiliated with each institute's OpenAlex ID (I26072440, I9747756, I181514455), 2016–2026; journal and conference papers only (preprints, book chapters, errata, retracted and duplicate versions removed; a version's citations are merged into the published paper). Journal quartile = Scimago SJR quartile of the journal in the publication year; conference rank = ICORE2026 (strict acronym + title match). Indian institutions = OpenAlex country code IN. *Core faculty* = authors with ≥5 papers in ≥3 years, last author on ≥30% and first author on ≤50% of them; departments come from 'Department of …' in the affiliation strings. 2026 is a partial year.

## 2. Publication trend and venue quality

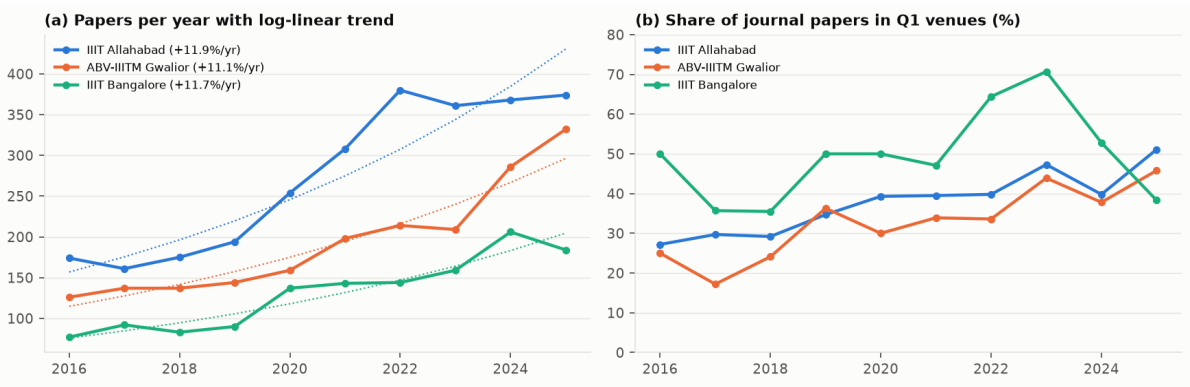



Figure 1. (a) Yearly output with fitted log-linear growth (2026 omitted as partial). (b) Share of journal papers in Q1 venues.

| Institution | Top 5 journals (papers, SJR quartile) |
|---|---|
| IIIT Allahabad | Multimedia Tools and Applications (56, Q1); Journal of Biomolecular Structure and Dynamics (44, Q2); Wireless Personal Communications (32, Q3); IEEE Access (30, Q1); IEEE Sensors Journal (24, Q1) |
| ABV-IIITM Gwalior | Multimedia Tools and Applications (28, Q1); International Journal of Communication Systems (23, Q2); Wireless Personal Communications (20, Q3); IEEE Internet of Things Journal (17, Q1); IEEE Access (17, Q1) |
| IIIT Bangalore | IEEE Access (11, Q1); IEEE Transactions on Communications (8, Q1); IEEE Communications Letters (7, Q1); International Journal of Computer Sciences and Engineering (7, –); IEEE Transactions on Wireless Communications (5, Q1) |

## 3. Faculty, departments and research areas

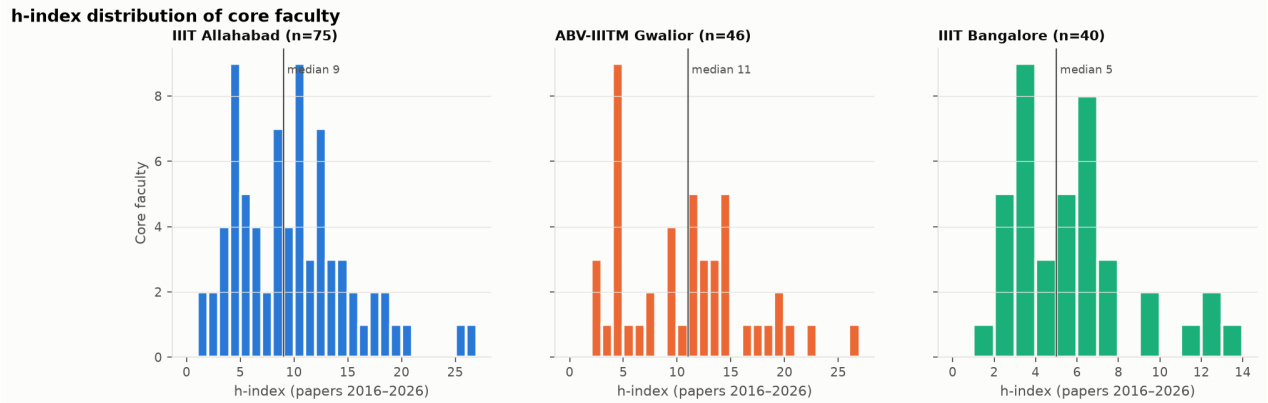



Figure 2. h-index (2016–2026 papers) of core faculty. Output and h-index are strongly related (Spearman ρ = 0.868); median OpenAlex lifetime h-index of core faculty = 13.

| Field (% of papers) | Computer Science | Engineering | Materials Science | Medicine | Business, Management and Accounting | Social Sciences |
|---|---|---|---|---|---|---|
| IIIT Allahabad | 39.2 | 24.8 | 5.5 | 4.6 | 2.9 | 1.4 |
| ABV-IIITM Gwalior | 36.9 | 25.7 | 7.5 | 6 | 7 | 2.9 |
| IIIT Bangalore | 49.5 | 25.5 | 0.7 | 2.6 | 1.7 | 6 |

Largest departments by papers — **IIIT Allahabad**: Information Technology (1152), Electronics & Electrical Engineering (753), Applied Sciences & Mathematics (728); **ABV-IIITM Gwalior**: Computer Science & Engineering (768), Electronics & Electrical Engineering (605), Applied Sciences & Mathematics (524); **IIIT Bangalore**: Computer Science & Engineering (832), Electronics & Electrical Engineering (597), Management & Humanities (148).

| Institution | Top 5 core faculty by h-index (papers, h, department) |
|---|---|
| IIIT Allahabad | Pritish Kumar Varadwaj (122, h=26, Applied Sciences); Sonali Agarwal (128, h=25, Information Technology); Shekhar Verma (113, h=20, Information Technology); Satish Kumar Singh (83, h=19, Computer Science); Sintu Kumar Samanta (54, h=18, Applied Sciences) |
| ABV-IIITM Gwalior | Anurag Srivastava (157, h=26, Applied Sciences); Shashikala Tapaswi (77, h=22, Computer Science); Manoj Kumar Dash (50, h=20, Management); Anupam Shukla (53, h=19, Computer Science); K. V. Arya (86, h=19, Computer Science) |
| IIIT Bangalore | Dinesh Babu Jayagopi (70, h=13, Computer Science); Shrisha Rao (49, h=12, Computer Science); Madhav Rao (161, h=12, Electronics); Debabrata Das (140, h=11, Electronics); Neelam Sinha (50, h=9, Electronics) |

## 4. Citation impact beyond raw counts

| # | Most-cited paper | Institution | Venue | Year | Cites | DOI |
|---|---|---|---|---|---|---|
| 1 | The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major… | ABV-IIITM Gwalior | The Astrophysical Journal | 2022 | 5,037 * | doi |
| 2 | Activation functions in deep learning: A comprehensive survey and benchmark | IIIT Allahabad | Neurocomputing | 2022 | 1,250 | doi |
| 3 | Green synthesis of metallic nanoparticles as effective alternatives to treat antibiotics resistant bacterial… | IIIT Allahabad | Biotechnology Reports | 2020 | 605 | doi |
| 4 | Structural insights into conformational stability of both wild-type and mutant EZH2 receptor | IIIT Allahabad | Scientific Reports | 2016 | 432 | doi |
| 5 | Linking Corporate Social Responsibility (CSR) and Organizational Performance: the moderating effect of… | IIIT Allahabad | European Research on Management and… | 2020 | 382 | doi |
| 6 | Energy-Efficient Intelligent Routing Scheme for IoT-Enabled WSNs | ABV-IIITM Gwalior | IEEE Internet of Things Journal | 2021 | 335 | doi |
| 7 | Graphene oxide: An efficient material and recent approach for biotechnological and biomedical applications | IIIT Allahabad | Materials Science and Engineering C | 2018 | 313 | doi |
| 8 | Exploring the sustainability performances of firms using environmental, social, and governance scores | ABV-IIITM Gwalior | Journal of Cleaner Production | 2019 | 307 | doi |
| 9 | Defect-Engineered Graphene for High-Energy- and High-Power-Density Supercapacitor Devices | ABV-IIITM Gwalior | Advanced Materials | 2016 | 303 | doi |
| 10 | Towards edge computing in intelligent manufacturing: Past, present and future | ABV-IIITM Gwalior | Journal of Manufacturing Systems | 2022 | 284 | doi |

* >50-author collaboration (Astropy, 100+ authors) – excluded from normalised scores and the regression.

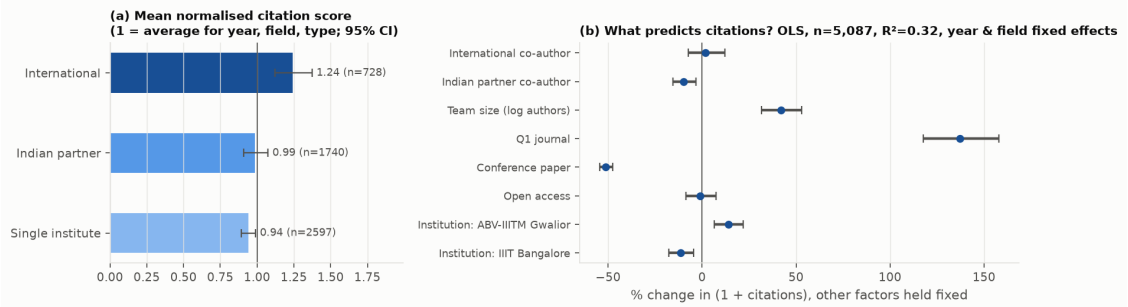



Figure 3. (a) Mean normalised citation score (citations ÷ mean of papers with the same year, field and type; papers up to 2024; bootstrap 95% CI). (b) OLS of log(1+citations) with year and field fixed effects and HC1 robust standard errors.

| Institution | Growth / yr (95% CI) | Q1 share 2016–19 → 2022–25 | Mean normalised impact (95% CI) | Papers in top 10% | Citation Gini | Top-10% papers' share |
|---|---|---|---|---|---|---|
| IIIT Allahabad | 11.9% (8.8–15.0) | 30.2% → 44.2% | 1.013 (0.944–1.088) | 9.2% | 0.733 | 58.6% |
| ABV-IIITM Gwalior | 11.1% (8.9–13.4) | 26.2% → 40.8% | 1.048 (0.982–1.119) | 11.9% | 0.778 | 63.4% |
| IIIT Bangalore | 11.7% (8.9–14.5) | 42.5% → 54.6% | 0.901 (0.828–0.98) | 7.5% | 0.728 | 57.9% |

## 5. Citation network

- **Construction.** Directed paper → paper edges from OpenAlex *referenced_works*: 6,543 papers, 5,946 internal edges (2,243 papers cited at least once, largest weak component 837). citation_network.csv also holds 21,069 out-flow and 18,931 in-flow edges with other Indian institutes.
- **In-degree** is heavy-tailed (distribution on the dashboard's citation-network page): 4,300 papers are never cited internally, 986 once, and the maximum is 25.
- **Self-citation and homophily.** 94.0% of internal edges share an author and 97.4% stay inside the citing institute (shuffled baseline 40.3% ± 0.59, z = 97.1). PageRank therefore ranks the most-continued research lines; it correlates only moderately with global citations (ρ = 0.531).

| # | PageRank top 10 (internal graph) | Institution | Year | In-deg | PageRank |
|---|---|---|---|---|---|
| 1 | Physical Layer Security in Cooperative AF Relaying Networks With Direct Links Over Mixed Rayleigh and… | IIIT Allahabad | 2018 | 25 | 0.00207 |
| 2 | ULTRA-WIDEBAND CPW FED BAND-NOTCHED MONOPOLE ANTENNA OPTIMIZATION USING MACHINE LEARNING | ABV-IIITM Gwalior | 2022 | 12 | 0.00201 |
| 3 | Quad-port ring dielectric resonator based MIMO radiator with polarization and space diversity | IIIT Allahabad | 2020 | 10 | 0.00200 |
| 4 | Noise robust face hallucination algorithm using local content prior based error shrunk nearest neighbors… | ABV-IIITM Gwalior | 2018 | 18 | 0.00190 |
| 5 | Inception U-Net Architecture for Semantic Segmentation to Identify Nuclei in Microscopy Cell Images | IIIT Allahabad | 2020 | 11 | 0.00179 |
| 6 | Investigation of Dual-Band Rectangular Dielectric Resonator Antenna with DGS for Wireless Applications | ABV-IIITM Gwalior | 2021 | 2 | 0.00176 |
| 7 | Defect-Engineered Graphene for High-Energy- and High-Power-Density Supercapacitor Devices | ABV-IIITM Gwalior | 2016 | 8 | 0.00169 |
| 8 | Dynamics of a prey-generalized predator system with disease in prey and gestation delay for predator | ABV-IIITM Gwalior | 2016 | 16 | 0.00152 |
| 9 | Circularly Polarized Quad-Port MIMO Dielectric Resonator Antenna with Beam Tilting Feature for Vehicular… | IIIT Allahabad | 2021 | 9 | 0.00151 |
| 10 | Photoacoustic field calculation for nonspherical axisymmetric fluid particles | IIIT Allahabad | 2016 | 12 | 0.00143 |

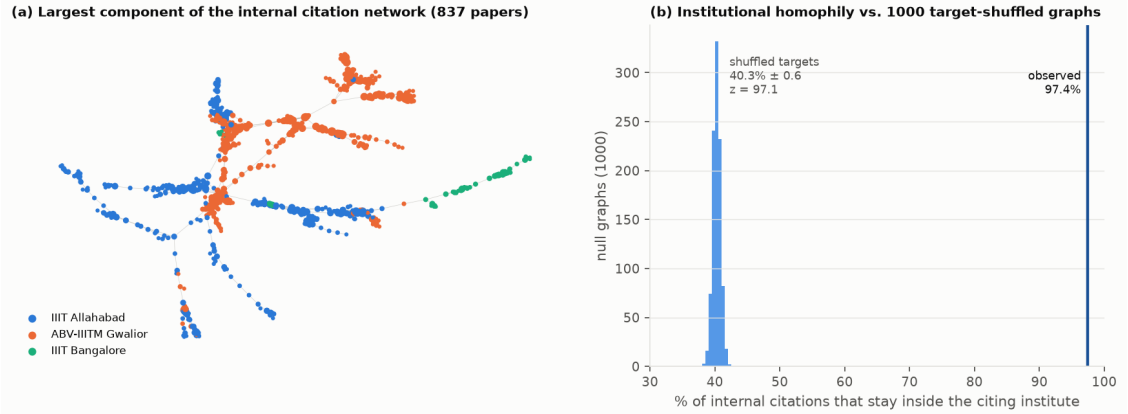



Figure 4. (a) Largest weakly connected component of the internal citation network, coloured by institute: the three institutes form separate regions. (b) Observed same-institute share versus 1000 graphs with shuffled citation targets.

## 6. Authorship network

- **Construction.** Undirected co-authorship among the 4,041 authors of the three institutes (OpenAlex author IDs; extra merges only with same name, same institute and a shared co-author — DECISIONS.md), 8,667 weighted edges; authorship_network.csv adds 8,589 edges to co-authors at other Indian institutes.
- **Structure.** Small-world (clustering 0.709 vs 0.0015, path length 6.08 vs 5.0; σ = 378.0), 532 Louvain communities with modularity 0.887 (configuration-model graphs: 0.469). Communities align with institutes (NMI 0.454) and departments (NMI 0.426); institute assortativity 0.974.
- **Centrality.** Core faculty are the hubs (median degree 19 vs 3). Bridges with the highest betweenness link otherwise separate groups.

| Degree centrality (co-authors) | Institution | Betweenness centrality | Institution |
|---|---|---|---|
| Madhav Rao (154) | IIIT Bangalore | Mahua Bhattacharya (0.134) | ABV-IIITM Gwalior |
| Om Prakash Vyas (113) | IIIT Allahabad | Madhav Rao (0.124) | IIIT Bangalore |
| Manish Goswami (97) | IIIT Allahabad | Santosh Singh Rathore (0.120) | ABV-IIITM Gwalior |
| Sonali Agarwal (92) | IIIT Allahabad | Daksh Sharma (0.110) | IIIT Bangalore |
| Shekhar Verma (84) | IIIT Allahabad | Meenakshi D'Souza (0.098) | IIIT Bangalore |
| Jyotsna L. Bapat (80) | IIIT Bangalore | Sonali Agarwal (0.077) | IIIT Allahabad |
| Kavindra Kandpal (78) | IIIT Allahabad | Ashutosh Kumar (0.070) | IIIT Allahabad |
| Pritish Kumar Varadwaj (74) | IIIT Allahabad | Pritish Kumar Varadwaj (0.068) | IIIT Allahabad |

## 6b. How the three institutes connect

| Pair | Co-author links (expected) | Co-published papers | Min. node cut | Mean distance | Shared Indian partners (Jaccard) | Non-self citations A→B / B→A |
|---|---|---|---|---|---|---|
| IIITA-ABV | 116 (2,232) | 29 | 28 | 5.72 | 178 (0.335) | 42 / 21 |
| IIITA-IIITB | 17 (2,073) | 8 | 8 | 8.39 | 93 (0.19) | 4 / 2 |
| ABV-IIITB | 5 (915) | 1 | 8 | 7.09 | 86 (0.209) | 9 / 7 |

- **The IIITA–ABV link is carried by people who moved, not by joint projects.** 28 of the 37 papers that link two IIITs include an author whom OpenAlex places at both institutes (35 such authors, 12 of them core faculty). The only two mixed communities are IIITA+ABV. *Caveat:* 'Placed at both' is inferred from affiliations over 2016–2026; it may be a move, a dual affiliation, or an OpenAlex affiliation error. We cannot tell these apart without CVs.

- **IIIT Bangalore hangs on eight people.** Mean shortest-path distance IIITA–IIITB = 8.39 and ABV–IIITB = 7.09 steps, vs 4.42 inside IIITB and 5.72 for IIITA–ABV. In citations, IIITB papers cite the other two only 9 times once author self-citations are removed. *Caveat:* Node cuts are computed on the 2016–2026 co-authorship snapshot; one new joint project could change them.
- **Cross-institute reachability collapses when brokers are removed.** Removing the same number of random authors leaves 62% and 60% (10 runs, sd ≤ 0.6 pp). The top 5% of authors hold 33.1% of all co-author ties; max degree 154 vs 13.4 in an ER graph of the same size. *Caveat:* Betweenness is computed once (non-adaptive removal); removing a person is a thought experiment, not a forecast.

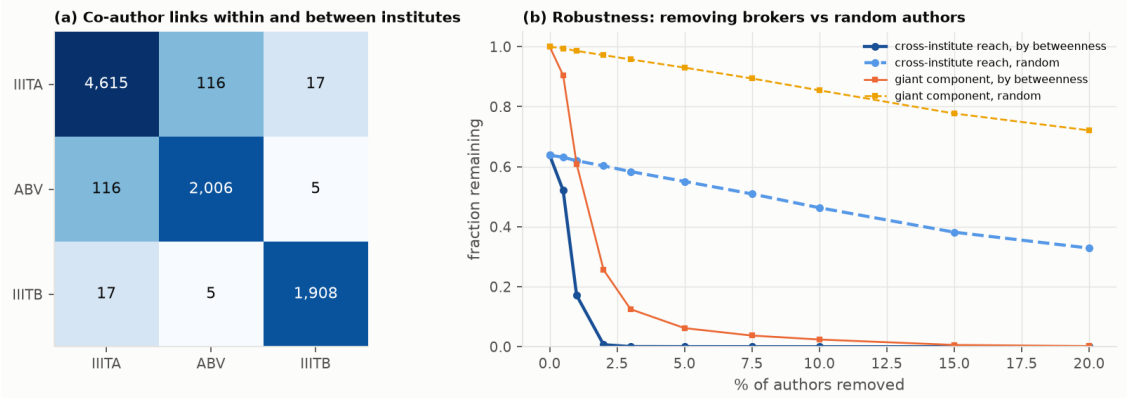



Figure 5. (a) Co-author links within (diagonal) and between institutes. (b) Share of cross-institute author pairs still connected, and giant-component size, when authors are removed by betweenness or at random. Within-institute distances: IIITA 4.69, ABV 4.24, IIITB 4.42 steps; network diameter 17.

## 7. Indian connectivity: citation flow and collaboration

| Institution | Cites received / given | Ratio | Top 5 Indian institutes citing us | Top 3 co-author partners |
|---|---|---|---|---|
| IIIT Allahabad | 10,779 / 11,519 | 0.936 | Saveetha University (419); Vellore Institute of Technology University (392); National Institute Of Technology Silchar (240); SRM Institute of Science and Technology (240); University of Allahabad (213) | Motilal Nehru National Institute of Technology (62, NIT); University of Allahabad (57, Central University); National Institute Of Technology Silchar (55, NIT) |
| ABV-IIITM Gwalior | 9,312 / 9,591 | 0.971 | GLA University (453); National Institute of Technology Patna (288); Vellore Institute of Technology University (276); Thapar Institute of Engineering & Technology (239); Chitkara University (193) | Jiwaji University (65, State/Private University); Indian Institute of Technology Kanpur (55, IIT); Indian Institute of Technology BHU (54, IIT) |
| IIIT Bangalore | 1,790 / 2,591 | 0.691 | Indian Institute of Science Bangalore (128); Amrita Vishwa Vidyapeetham (56); Vellore Institute of Technology University (46); SRM Institute of Science and Technology (41); Malaviya National Institute of Technology Jaipur (40) | Samsung (India) (79, Industry); Indian Institute of Science Bangalore (46, IISER/IISc/NISER); National Institute of Mental Health and Neurosciences (36, Medical) |

Co-authorship by partner cluster (shared papers, all three institutes): State/Private University 1795, IIT 643, NIT 384, IIIT 334, Central University 282, Govt/Other Research 144, Industry 140, CSIR 102. Cross-cluster links (partner not an IIIT) dominate. Topic similarity (Jaccard over OpenAlex subfields) is highest between IIITA and IIITB (0.5) and with IIIT Hyderabad for IIITB, yet it does not predict collaboration volume.

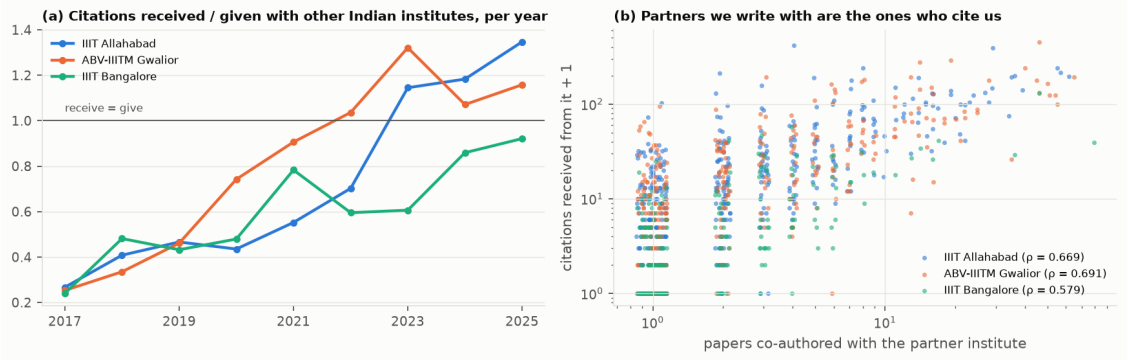



Figure 6. (a) Yearly ratio of citations received from / given to other Indian institutes. (b) Co-authored papers vs citations received per partner institute.

## 8. Limitations

- OpenAlex affiliation and reference coverage is incomplete, especially before 2018. Author identity trusts OpenAlex IDs and merges further only with co-author evidence, so some people may remain split; with a name-only rule there would be 264 instead of 138 inter-institute links. The core-faculty rule is a heuristic, not an official faculty list.
- Networks are 2016–2026 snapshots; node cuts and robustness describe that snapshot, and an author's 'main' institute is where most of their papers are.
- IIIT Bangalore authors rarely state a department, so most of its departments are inferred from the research field; CORE ranks cover only venues on the ICORE list.
- Citation counts are taken at collection time; 2025–26 papers are excluded from normalised scores. The regression shows associations, not causal effects.

## 9. Conclusions

- All three IIITs roughly doubled their output in ten years while moving to better journals, so the growth is not a volume-only effect. Differences in raw citations between them largely disappear once papers are compared with others of the same year, field and type.
- Citation success depends far more on where a paper is published (Q1 journal vs conference) and how large the team is than on whether it has foreign co-authors.
- Both networks show three largely separate research systems: internal citations are almost all self-citations, and only 37 papers connect two of the IIITs — 28 of them through people placed at two institutes. IIITB is attached to the others through 8 authors. Their Indian links run through NITs, IITs and nearby universities, and partners that co-author with an institute supply most of its Indian citations (57.0–72.5%).
- Implication: joint IIIT projects and targeted partnerships are the clearest lever for visibility; topic overlap alone does not create collaboration.

## 10. Dashboard guide

**All institutions**: Publication trend · quartile (Q1–Q4) · CORE rank · research areas · top-10 cited papers with DOIs · **IIIT Allahabad / ABV-IIITM Gwalior / IIIT Bangalore**: Faculty list (name, department, h-index, papers) · sortable performance table · h-index histogram · top-5 journals and conferences · departments · **Citation network**: Directed citation graph (D3) · PageRank top-10 · in-degree distribution · **Network explorer**: Interactive co-authorship graph (filters, search, author details, Louvain colouring) · **Network analysis**: Network properties · log-binned degree distribution · centralities · communities · robustness · null models · **Institution comparison**: Per-institute structure · links, distances and citations between institutes · bridges · node cuts · **Indian connectivity**: Citation in/out-flow per year · impact ratio · top-10 citing institutes and papers · partner ranking with cross-cluster flags · Jaccard similarity · **Findings & modelling**: Structural findings · growth and Q1 tests · normalised impact · regression · methods · limitations · **Course connections**: Degree & paths · log-binning · Erd■s–Rényi · ring/star path lengths

## 11. Individual contributions

| Member | Work package | Difficulty | Learning | Hours | Effort |
|---|---|---|---|---|---|
| Hrishikesh | OpenAlex collection under the 1000-request/day limit: batched cites / cited_by filters, page paging, resumable caches | 4 | 4 | 14 | 56 |
| Hrishikesh | Corpus rules: journal/conference classification, version merging, de-duplication, 100-author truncation fix | 3 | 3 | 8 | 24 |
| Hrishikesh | Year-matched Scimago quartiles and strict ICORE2026 matching | 3 | 3 | 7 | 21 |
| Hrishikesh | Data validation: flag checks, cluster labels, partner-ID alignment, folder clean-up | 2 | 2 | 6 | 12 |
| Sarvesh | Citation network: internal / in-flow / out-flow edges, PageRank, in-degree, self-citation and homophily null model | 4 | 4 | 12 | 48 |
| Sarvesh | Authorship network: author disambiguation, Louvain, exact betweenness, small-world and configuration-model tests | 4 | 4 | 11 | 44 |
| Sarvesh | Indian connectivity: citation flow, impact ratio, partner clusters, Jaccard topic similarity | 3 | 3 | 7 | 21 |
| Shourya | Faculty and department identification, core-faculty rule, h-indices | 3 | 3 | 8 | 24 |
| Shourya | Statistical inference: normalised impact, OLS with robust errors, trend and concentration tests | 4 | 4 | 10 | 40 |
| Shourya | Interactive dashboards (D3 / Chart.js and Streamlit) and report figures | 3 | 3 | 12 | 36 |
| Shourya | Report writing and layout | 2 | 2 | 6 | 12 |

Effort = hours × mean(difficulty, learning curve), both on a 1–5 scale. Totals: Hrishikesh 113 (33.4%), Sarvesh 113 (33.4%), Shourya 112 (33.1%) — an equal three-way split. Every part was discussed, cross-checked and reviewed by all three members.